

# Andrei Ovsiannikov

## Full Stack Developer

### 10+ years of experience

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## About

I specialize in designing project architecture and building scalable, high-performance systems using TypeScript(Node.js, NestJS, Express), Python(FastAPI, Django), and modern front-end frameworks (Vue/Nuxt, React/Next, Angular). I'm also proficient in PostgreSQL, MongoDB, Docker, Nginx, Redis, RabbitMQ, and Kafka. I have experience leading projects, integrating machine learning, and optimizing performance.

## Degree

Artificial intelligence – Master of Computer Science (MCS) – 2018 – SibSAU.
Software engineering – Bachelor of Computer Science (BCompSc) – 2016 – SibSAU.

## Skills

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| **Programming languages** | JavaScript, TypeScript, Python, C#, Go |
| **Front end** | Vue, Nuxt, React, Nextjs, Angular, Svelte, HTML, CSS, Bootstrap, Material, Tailwind, Sass, Less |
| **Back end** | NodeJS, Express, Nestjs, ElysiaJS, FastAPI, Django/DRF, Flask |
| **Cloud** | AWS (Amplify, Lambda, S3, Cognito, CloudWatch), GCP, Azure |
| **Operating Systems** | Linux, Windows |
| **DBMS/MQTT/ORM** | PostgreSQL, MongoDB, MySQL, Redis, RabbitMQ, Kafka, BullMQ, DynamoDB, Prisma, Drizzle, TypeORM, Sequalize, SqlAlchemy |
| **Other technologies** | Docker, Kubernetes, Helm, Nginx, Git, WebRTC, SocketIO, Anthropic, ChatGPT, Gemini, phi3, Llama, LlamaIndex, LangChain, ChromaDb, Langflow, Airflow, RAGAS, Optuna, Cohere, OpenTelemetry, Grafana, Prometheus, gRPC, Terraform |
| **Languages** | Russian (Native), English (B2) |

## Professional experience

**05/2025 to date – layer3.press**
**Full-stack Developer**

**Skills:** JS/TS(Bun, Elysia, Drizzle, React, Remix, Zustand), PostgreSQL, Docker, Kubernetes, AWS, Kafka, Redis, Nostr, LLMs(ChatGPT, Gemini).

**Developed** layer3.press, a multi-tenant blogging platform with email and Nostr subscriptions. Designed and implemented core platform architecture, including backend APIs, frontend applications, event-driven services, and scheduled workflows. Built authentication, publishing, and content management systems with email OTP and Nostr authentication via secret keys or browser extensions. Developed a Nostr blogging system using Nostr protocol specifications (NIPs), event-driven architecture, multi-relay publishing, event indexing and caching, profile synchronization, and automated publishing workflows. Created content management tools including a customizable MJML email template editor and a Lexical editor with custom plugins for images, YouTube embeds, Nostr integration, and paywall functionality. Implemented

Markdown-to-HTML conversion pipelines. Developed subscription and monetization systems with Stripe Connect integration, paid subscriptions, paywalled content access, configurable subscription plans, publisher membership management, and gamified onboarding flows with automated publishing protection. Implemented AI content processing and generation pipelines including summarization, translation, image generation, content generation from crawled publisher data, opinion analysis, real-time narrative generation, and AI article re-ranking. Built automated content generation and publishing infrastructure with RSS/site crawling pipelines, RSS normalization, AWS KMS secure key management, high-load background job processing with SSE notifications, X (Twitter) synchronization, and hCaptcha-protected forms.

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**09/2023 to 04/2025 –** "EURO ATLANTIC SMART TECHNOLOGY LTD (SP. Z O.O)" (Poland)
**Tech Lead, AI/ML Engineer, Full-stack Developer**

**Skills:** JS/TS(Nestjs, Prisma), Python(FastAPI, PyTorch, nltk), Azure (OpenAI, Storage, AI AzureSpeach, Custom question answering), LLM(Anthropic, ChatGPT, Gemini, phi3, Llama, LlamaIndex), LangChain, ChromaDb, PostgreSQL, Docker, Kubernetes, AWS(S3, transcribe, EC2), RabbitMQ, Redis, Firebase.

**Designed and implemented** AI Q&A chat system (REST API + SSE) with integration of modern LLMs and cloud services, and knowledge base management platform with OCR, audio transcription, pre-processing for RAG systems and integration with LMS Docebo, Google Drive, AWS. Conducted in-depth R&D into RAG quality improvement, systematically analyzing document preprocessing and contextual enrichment, retrieval and query optimization, entity/relationship extraction, hybrid search, conversation context, context compression and emerging approaches including Graph RAG, Corrective RAG, Self-Reflective RAG and HyDE. Improved retrieval quality using RAGAS, Optuna and Cohere re-ranking. Led integration of LangChain and Langflow for improved AI workflows.

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**11/2022 to 09/2023 – UDev – Full-stack Developer**

**Skills:** JS/TS(NestJs, Prisma), Python(FastAPI, Django), SocketIO, React, PostgreSQL, Docker, Kubernetes, Kafka, MongoDB, GCP.

1. **Migrated** the Tumeke (**https://www.tumeke.io/**) project from Python to NestJS; redesigned its architecture for optimization; developed an algorithm to calculate and visualize Ergonomic Posture Risk Assessment; optimized raw SQL queries; improved response times by 30x.
2. **Developed** smart speaker skills to play background music using NLP command processing.

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**03/2022 to 11/2022 – Freelance(Upwork) – Full-stack Developer**

**Skills:** TS, Node(Express), Python(Flask, Tornado), Vue, React, GraphQL, RabbitMQ, Docker, PostgreSQL.

**Developed** a tournament scheduler for the gamified assessment system Cibirlan; algorithms for team creation with role assignment; real-time chat using Socket.IO.

**Implemented** an admin panel for a video surveillance system with a customizable dashboard, widgets, and WebRTC pipelines.

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**07/2021 to 03/2022 – Altermeliora – Full-stack Developer**

**Skills:** TS, NodeJS(NestJS), React.js, Python(Django), GCP, PostgreSQL, Redis, Docker, MongoDB.

**Developed** a payment platform for selling courses and physical goods. Implemented product form; category hierarchy builder; invoice form with payment system settings, discount and promo code logic; customizable invoice flow with auto-registration on third-party services; sales reports; tax calculation for sellers. Integrated delivery services (Shiptor,ChinaDivision); payment systems (Bluesnap,Fondy,PayPal,YooMoney,Stripe); email delivery (SendGrid); external course-selling platforms. Built calorie calculation, cross-platform sync, and multilingual interfaces.

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**10/2020 to 06/2021 – Freelance(Upwork) – Full-stack Developer**

**Skills:** TS, Node(Nest.js), Angular, AWS(Amplify/Lambda/DynamoDB/Cognito), RabbitMQ, Docker, PostgreSQL, MongoDB.

1. **Developed** an admin panel for the MVP of a freight transportation system with Cognito authorization, geocoding and data visualization using here.com.

2. **Designed and implemented** scalable architecture for white-label licensing for I-EXP projects. Built a Telegram bot, neural network training data generation tools, image similarity vector search, and authentication.
3. **Built** a website security analysis system.

**08/2018 to 10/2020 – YLab – Lead Full-Stack Developer**

**Skills:** Python(FastAPI,Django/DRF,Flask), SqlAlchemy, TensorFlow, TypeScript, AngularJS, Vuejs, Go, PostgreSQL, Redis, Celery, Docker, Minio, S3, scrapy/selenium, Clickhouse, MSSql.

1. onfin.io (ECN.Broker): **Led development of a fintech trading platform** including admin panel, user account, trader training pages, landing pages, email templates, authorization form. Built microservice for *MetaTrader 4* API, Telegram bot, referral program, forex calcs. Integrated payment systems (*RBK.money*, *Accentpay, Interkassa, SorexPay)*; *WYSIWYG editor Quill*; *Claws&Horns)*; *google visualization charts*; *IndigoSoft*; *RAMM copy trading*; *Bpilot* for contract specification; *SendGrid* for email delivery.
2. *B2B-export*: **Built** admin panel, user account pages, email templates, category hierarchy builder, bulk product import. Integrated ABBYY API for product translations.
3. *Carreta*: **Implemented** product prices crawler and bulk price import feature.
4. **Developed internal projects**: CSV/XLS report aggregator, user testing system, OCR tool.

**09/2016 to 02/2018 – Aigeo – Full-stack Developer**

**Skills:** Python(Django/SQLAlchemy), TypeScript, AngularJS, OpenLayers 3, PostgreSQL.
**Developed** geospatial tools and applications for Yenisei GIS mapping platform. Focused on medical GIS features, combining OpenLayers with backend data pipelines using Python and PostgreSQL with PostGIS. Implemented Aigeo References, a comprehensive geospatial reference system. Created normalization tools to improve the accuracy of mapping algorithms.

## Pet Project - MMOSG Game ManeTalk

**Skills:** NestJS, Vue.js, Nuxt, Electron, RabbitMQ, Kafka, Docker, Godot, PostgreSQL, Minio.
**Developed** the **ManeTalk** website with SSO authentication and backend for game modules. Designed scalable REST and event-driven APIs for managing profile data, character textures, inventories, and real-time chat, integrated with the Godot Engine. Implemented DDoS and bot protection via per-endpoint rate limiting, gibberish and Detoxify message classification, and multi account spam/flood detection using Longest Common Subsequence analysis. Built a Telegram Bot for admin notifications and moderation, and a desktop game launcher with a built-in anti-cheat system.